Yingming Zhang

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Research Interests: Soft robotics; cable-driven and shape-morphing structures; actuator design and inverse design; mechanics modeling and simulation.

Education

Purdue University, West Lafayette, IN, USA **Aug. 2025 – Present**
M.S., Engineering Technology | GPA: 3.86/4.0 | Expected graduation: Spring 2027

Huazhong University of Science & Technology (HUST), Wuhan, China **Sept. 2011 – Jun. 2018**
M.Eng., Materials Processing Engineering | GPA: 3.65/4.0 | Sept. 2015 – Jun. 2018
B.Eng., Material Forming and Control Engineering | GPA: 3.82/4.0 | Minor in Business Administration (GPA: 3.61/4.0)

Research Experience

Cable-Driven Shape Morphing and Inverse Design of Flexible Membranes **Sept. 2026 – Present**
Graduate Research, Purdue University

- Addressing an actuator-design problem: achieving complex target shapes in an initially flat membrane with a minimal number of linear actuators, where each actuator drives coupled displacements at multiple locations.
- Modeled the system in COMSOL and Abaqus, capturing the perforation geometry and cable–membrane constraints that govern how actuation is transmitted across the membrane, and compared both approaches for accuracy and efficiency.
- Developed a simplified steady-state model with modified cable–membrane constraints (**~6 min vs. ~1 h per run** for the full transient model) to support parametric studies of actuator configurations.
- Building a simulation dataset (Python, NumPy) that maps actuator configurations to deformed shapes as the basis for data-driven inverse design.

Woven Architected Structures: Point-Cloud Reconstruction and Deformation Analysis **May 2026 – Aug. 2026**
Summer Research, University of Illinois Urbana–Champaign

- Addressed a measurement gap: the internal deformation of spatially complex woven (interlaced network) structures is inaccessible to surface-based measurement techniques.
- Developed Python algorithms to reconstruct complex 3D woven geometries from X-ray CT point clouds and convert them into CAD models, making the hidden internal architecture **quantitatively accessible**.
- Developed point-cloud registration algorithms to align reconstructions before and after loading, enabling non-destructive analysis of load-induced internal deformation.

Aluminum Alloys and Advanced Manufacturing **Aug. 2025 – May 2026**
Research Assistant, Purdue University

- Combined CALPHAD phase calculations, HPDC simulations, and casting experiments to study fluidity and flow arrest in Al–Si–Cu–Fe alloys (Pub. [7]).
- First-authored a review in **International Materials Reviews (IF 17)** on processing–structure–property relationships in LPBF aluminum matrix composites (Pubs. [2], [3]).
- Prepared and cut specimens and performed material characterization, including metallography, optical microscopy, and tensile and hardness testing.

CAD Automation and Thermo-Mechanical CAE for Injection Molding **Sept. 2015 – Jun. 2018**
M.Eng. Research, Huazhong University of Science & Technology

- Developed algorithms and CAD tools (C++/C#, Siemens NX) for automatic generation of venting systems on complex freeform mold surfaces and conformal cooling channels (Pubs. [1], [5]).
- Performed thermo-mechanical CAE to evaluate mold deformation, part quality, and manufacturing defects.

Publications

Journal Articles

[1] **Zhang, Y.**, et al. Automatic generation of venting system on complex surfaces of injection mold. *The International Journal of Advanced Manufacturing Technology*, 2018, 98: 1379–1389.

[2] **Zhang, Y.**, et al. The laser powder bed fusion of particle-reinforced aluminum alloy matrix composites. *International Materials Reviews*, 2026. DOI: 10.1177/09506608261502610. **[IF: 17, JCR Q1]**

[3] **Zhang, Y.**, et al. Microstructure-controlled strengthening mechanisms and strategies in additively manufactured aluminum alloys: A review. *Engineering Science in Additive Manufacturing*, 2026, 2(1): 025490033.

[4] Li, M., **Zhang, Y.**, et al. Mechanical properties of γ-graphyne nanotubes. *RSC Advances*, 2018, 8(28): 15659–15666. [IF: 4.6]

Conference Papers

[5] **Zhang, Y.**, et al. Automatic design of conformal cooling channels in injection molding tooling. *IOP Conference Series: Materials Science and Engineering*, 2018, 307(1): 012025.

[6] Yang, Y., **Zhang, Y.**, Li, X. An improved artificial potential field algorithm based on nonuniform cell decomposition. *6th International Conference on Measurement, Instrumentation and Automation (ICMIA 2017)*, Atlantis Press, 2017: 735–739.

[7] **Zhang, Y.**, et al. A four-path framework for interpreting fluidity and flow arrest behavior in Al–Si–Cu–Fe die-casting alloys. *2026 NADCA Die Casting Congress & Tabletop*, 2026 (accepted).

Under Review

[8] He, X., Li, H., **Zhang, Y.**, et al. Corrosion behavior and failure analysis of laser powder bed fusion (LPBF) 18Ni300 steel die for high-pressure die casting. *Metallurgical and Materials Transactions B*, under review.

Professional Experience

Investment Manager (Technical Due Diligence), Cinda Investment Co., Ltd., China **Jul. 2018 – Jul. 2025**

- Conducted technical due diligence on advanced materials and manufacturing companies, assessing technology readiness, manufacturability, and scale-up risk.
- Translated laboratory-scale results into risk assessments and recommendations for investment committees.

Skills

Simulation: COMSOL Multiphysics, Abaqus, Siemens NX (CAD/CAE), Flow-3D, JMatPro (CALPHAD)
Programming: Python (NumPy, PyTorch), C++/C#
Imaging & Geometry: X-ray CT, point-cloud reconstruction and registration, ImageJ/Fiji, Blender
Characterization: Specimen preparation and cutting, metallography, optical microscopy, tensile and hardness testing